

Introduction to Representation

The segmentation technique is a decomposition of an image generally referred to as raw data, in the form of pixels along a boundary contained in a region.

Although these data sometimes are used directly to obtain descriptors that compact the data into representation that are considerably more useful in the computation of descriptors.

So, representation refers to the dealing of naming for segmented image dedicated to the object point.

Representation can be classified into two types:

- (i) Chain Codes
- (ii) Signatures

✓ 1) Chain Codes:

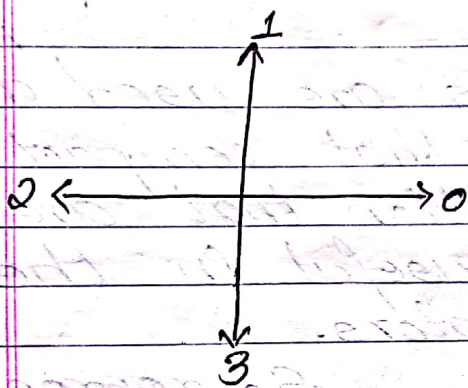
Chain codes are used to represent a boundary by a connected sequence of straight line segments of specified length and direction.

Typically, this representation

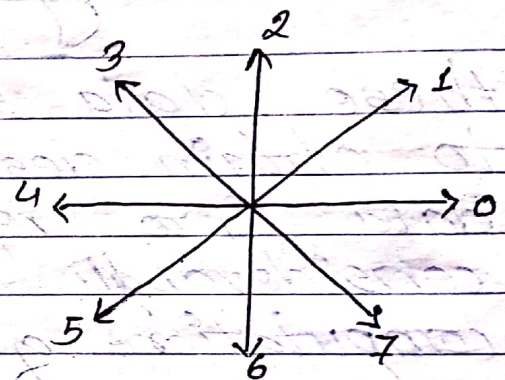
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Is based on four or eight (4 or 8) connectivity of the segments. The direction of each segment is coded by using a number.

Chain code on this scheme are referred as:



(fig (a): Direction number for 4 directional chain code)



(fig (b): Direction number for 8 directional chain code)

The chain code of a boundary depends on the starting point. However, the code can be normalized with respect to the starting point by treating it as a circular sequence of direction numbers & redefining the starting point so that the resulting sequence of numbers is a integer magnitude.

We can normalize by using rotation i.e 90°

45° for the codes using the difference of the chain code. Above figure shows the best example of 4 and 8 directional chain code.

2) Signatures:

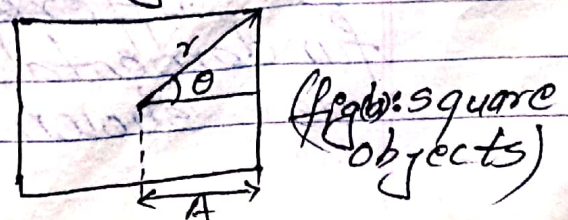
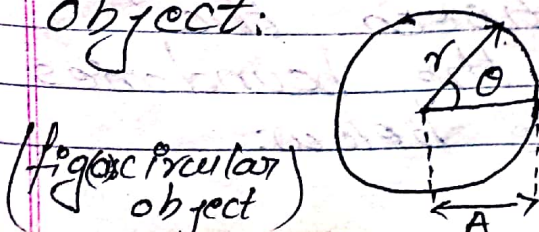
Signature is a 1-D functional representation of a boundary and may be generated in various ways.

One of the simplest method is plot the distance from an centroid point to the boundary as a function of angle.

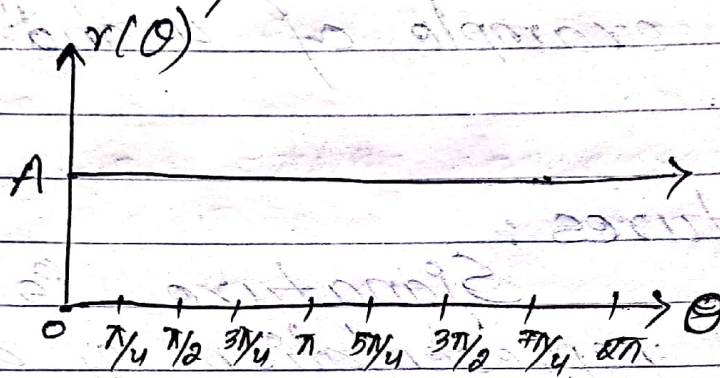
Signature generated by the approach just described are invariant to translation but they depend on rotation and scaling.

Normalization with respect to the rotation can be achieved by finding a way to select the same starting point to generate the signature.

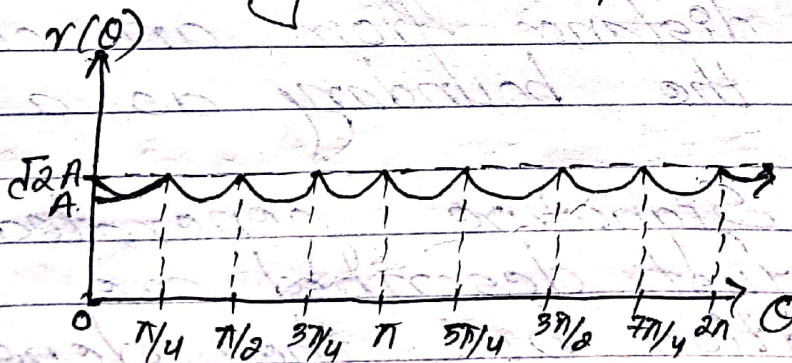
The signature can be defined by using the following object:



The corresponding angle signature can be defined as:

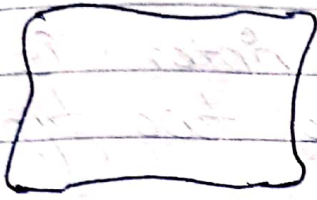


(fig (c): corresponding distance r vs angle signatures of circular objects)

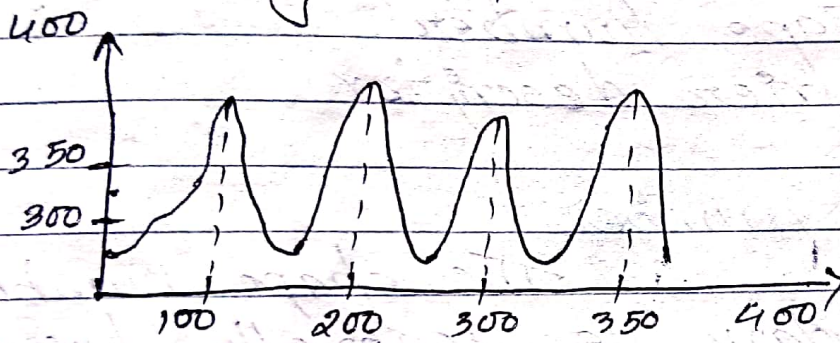


(fig (d): corresponding distance r vs angle signature of square objects)

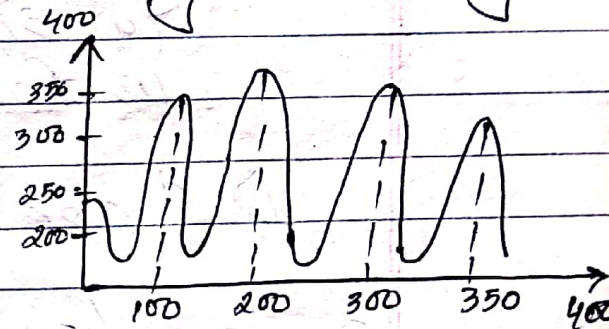
A single similar set of commands yielded the plot. Simply counting the number of prominent peaks in the signature is sufficient to differentiate between the fundamental set of the boundaries as shown in figure below:



(fig(a): Boundaries of an irregular square)



(fig(b): Boundaries of an irregular triangle)



(fig(c): Corresponding signature of an irregular square)

(fig(d): Corresponding signature of an irregular triangle)

Introduction to descriptors

Descriptors are also called features of an image that are useful when working with region boundaries.

Many of the descriptors are applicable to region and grouping of descriptors in the toolbox. The length of a boundary is one of its simplest descriptors.

The length of a four connected boundary is defined as the

number of pixel - 1. Descriptor is usually categorized into two types based on the boundary.

There are following types of descriptions:

- (a) Shape Number
- (b) Fourier descriptor

1) Shape number

The shape number of a boundary is generally based on four directional chain codes.

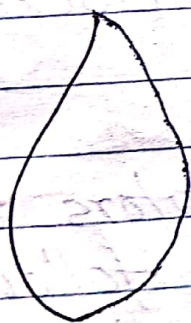
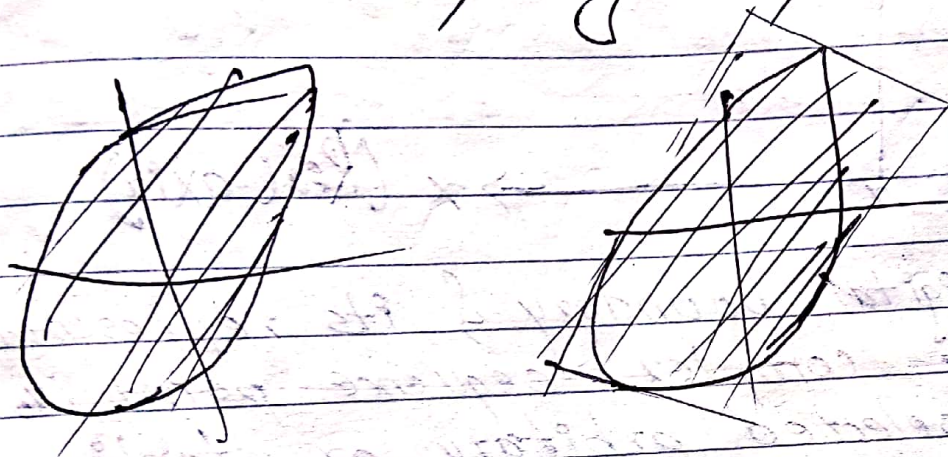
It is defined as the first difference Δ of a chain code of the smallest magnitude. The order n of a shape number is defined as the number of digits in its representation.

Thus, the shape number of a boundary is given by the parameter and order of shape number (length).

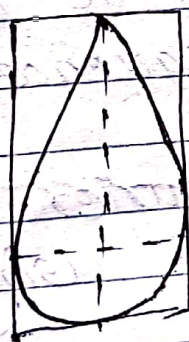
The 4th directional chain codes can be made insensitive to the starting point by using the integer of minimum magnitude & made rotation that are multiples of 90° by using the first difference of the code.

Thus, shape numbers are insensitive to the starting point & rotation that are multiple of 90° .

Shape number is illustrated step by step as:



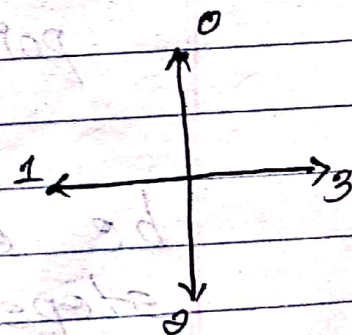
(i)



(ii)



(iii)



(iv)

Chain code = 000030032232201211
 Difference = 30003131131100310310
 Shape number = 000310330130031303

Fig: steps in the generation of shape number

2) Fourier description

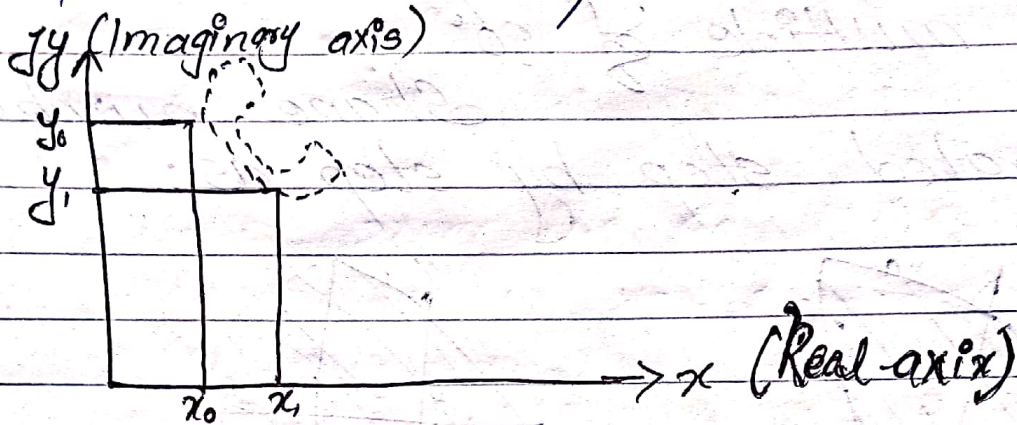


fig. A digital boundary & its representation as a complex sequence point (x_0, y_0) is selected arbitrary or starting point. Point (x_1, y_1) is the counter clockwise point in a sequence.

Fourier descriptors can be explained by using the following steps:

(i) Define the arbitrary point (x_0, y_0) & co-ordinate point or pairs $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_{k-1}, y_{k-1})$ are encountered in traversing the boundary in counter clockwise direction.

(ii) These co-ordinates can be expressed in the form:
 $x(k) = x_k$ and $y(k) = y_k$
 and

The boundary can be represented as the sequence of co-ordinates as:

$$S(k) = [x(k), y(k)], \text{ for } k = 0, 1, 2, \dots, k-1$$

The each co-ordinate pair can be treated as a complex number so that

$$S(k) = x(k) + jy(k)$$

(i) The discrete Fourier transform of 1D sequence, $S(k)$ can be written as

$$a(u) = \sum_{k=0}^{k-1} S(k) \cdot e^{-j \frac{2\pi u k}{K}}$$

where,

$$u = 0, 1, 2, \dots, K-1$$

(ii) The complex coefficient $a(u)$ are called "the Fourier descriptors" of the boundary. The inverse Fourier transform of these coefficients as:

$$S(k) = \frac{1}{K} \sum_{u=0}^{K-1} a(u) e^{j \frac{2\pi u k}{K}}$$

(iii) Therefore, for boundary descriptors, the Fourier descriptors can be explained as Fourier coefficients that are defined

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as:

$$\hat{S}(k) = \frac{1}{P} \sum_{u=0}^{P-1} a(u) e^{j \frac{2\pi U_k}{K}}$$

where,

P = preceding eqⁿ for $a(u)$ ~~where $P=$~~
where $P=0, 1, 2, \dots, K-1$

$\hat{S}(k)$ = Approximation to $S(k)$